

PRATHAM VISHWAKARMA

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SUMMARY

Software Development Engineer with 2+ years of experience specializing in backend development, Kubernetes orchestration, and CI/CD automation. Currently at Jio Platforms architecting Python/FastAPI automation systems and integrating Generative AI with infrastructure operations (MCP). Proven ability to build highly available, scalable cloud-native environments.

TOOLS & TECHNOLOGIES

Languages & Backend: Python, FastAPI, Shell Scripting

DevOps & Cloud: Docker, Kubernetes, Jenkins, ArgoCD (GitOps), Terraform, AWS, GCP

Observability & Tools: Prometheus, Grafana, SaltStack, Linux, Git, GitHub, Azure Repos

PROFESSIONAL EXPERIENCE

Software Development Engineer I | Jio Platforms Limited **Navi Mumbai, MH | Dec 2023 – Present**
Product Engineering – Jio CloudXP / Hybrid Cloud Management Platform

- **AI Infrastructure Orchestration (MCP):** Developed and exposed 50+ modular infrastructure tools (e.g., service status monitoring, Salt state execution, consumer lag tracking) using FastAPI, enabling AI agents to orchestrate complex infrastructure. Integrated Piko reverse tunnels and a custom AuthZ layer to securely execute LLM-triggered tool actions via the Salt Master API.
- **Cross-Platform Automation:** Engineered modular automation workflows using Python and Robot Framework. Implemented dynamic execution logic across Linux (SSH/SCP) and Windows (WinRM), increasing automation delivery speed by 60% and reducing manual execution by 50%.
- **System Optimization:** Designed OS-detection logic and secure, centralized storage upload pipelines with clickable UI retrieval. This cut file retrieval time by 80% and reduced manual post-run handling by 60%.

KEY PROJECTS

Notes-Studio: Cloud-Native Microservices Architecture **Dec 2025 – Mar 2026**
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- **Architecture & Kubernetes:** Engineered a containerized 3-tier architecture (FastAPI, PostgreSQL, Next.js) with JWT authentication. Orchestrated on Kubernetes using PVCs and Secrets for secure data persistence.
- **CI/CD & Observability:** Configured automated, zero-downtime deployment pipelines via Jenkins and integrated a full Prometheus & Grafana stack for real-time API and infrastructure monitoring.

E-Commerce Web Application (GitOps Deployment) **Jun 2025 – July 2025**

- **GitOps Automation:** Automated 95% of the deployment lifecycle using ArgoCD, improving release frequency by 70%. Maintained 99.9% uptime and reduced AWS EKS infrastructure costs by 25% through optimized scaling and Terraform provisioning.

EDUCATION

Bachelor of Technology – Computer Science **2019 – 2023**
Acropolis Institute of Technology and Research, Indore